Course Path

STAT 109 — Summer Session 1, 2026 (online, async)

This page shows the recommended path through the course during the five-week summer session from June 1 to July 2, 2026. You may work at your own pace within each module window, but the modules should be done in the order shown. Check Canvas to turn in work, take quizzes, see grades, and join discussions.

Reference sheets (methods map, R reference) summarize all course content. All readings also live on the lectures index if you want the full list.

The Supplemental (optional) column links extra textbook chapters (Vu & Harrington, free online) and videos/tutorials that match our Colab + `{tidyverse}` workflow — see Resources. None of these are required for the module project unless a row says otherwise.

How the course is graded

Component	Weight	What counts
Class participation	**5%**	Class Chatter (one per module, 0–5) + Q+A Discussion on Canvas
Quizzes	**25%**	All Canvas quizzes (Quiz 0/0B, Quiz A, Quiz B, synthesis per module) · best of 4 attempts per quiz · open notes
R coding exercise labs	**20%**	Submit exercise lab each module · graded on completion
Projects 1–5	**50%**	One-page summary + runnable R notebook each module
Total	**100%**	

At a glance

Module	Dates	Project due (recommended)
0 — Welcome	Jun 1-2	—
1 — Testing a Claim	Jun 2–8	Mon Jun 8
2 — Estimating the Truth	Jun 9–15	Mon Jun 15
3 — Discovering Patterns	Jun 16–22	Mon Jun 22
4 — Designing Fair Comparisons	Jun 23–29	Mon Jun 29
5 — Looking Ahead	Jun 30–Jul 2	Thu Jul 2

Canvas due dates are recommendations for pacing; all work to bee graded must be submitted by 11:59 pm PST July 2, 2026.

Module 0 — Start here

Step	Materials	Supplemental (optional)
Read 0	Module 0 Read 0	—
Read	Syllabus · Course path · Resources	—
Quiz	Quiz 0 (Start Here) · Quiz 0B (Question Ideas)	—

Step	Materials	Supplemental (optional)
Lab	Demo lab · Exercise lab	—
Canvas	Module 0 discussion (introduce yourself)	—
Supplemental pack	—	Colab tutorial · Posit: Getting Started with R · Textbook (free PDF) · Posit cheatsheets

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Module 1 — Testing a Claim

Is this observation surprising?

Step	Materials	Supplemental (optional)
Read 0	Module 1 Read 0	—
Read	Module 1 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 1 Read 2	—
Quiz B	Concept Check	—
Demo lab	Module 1 demo · Conditional demo	—
Exercise lab	Module 1 exercise lab	—
R reference	R quick reference	—
Milestone	Project 1 milestone · notebook	—
Synthesis	Module 1 synthesis	—
Project	Project 1	—
Supplemental pack	—	Textbook Ch 2 — Probability · Ch 3.2 — Binomial · Khan: binomial distribution · Khan: visualizing binomial

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Module 2 — Estimating the Truth

Building high-quality estimates from samples.

Step	Materials	Supplemental (optional)
Read 0	Module 2 Read 0	—
Read	Module 2 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 2 Read 2	Textbook Ch 4.2 — Confidence intervals
Quiz B	Concept Check	—
Demo lab	Module 2 demo	—
Exercise lab	Module 2 exercise lab	—
Milestone	Project 2 milestone	—
Synthesis	Module 2 synthesis	—
Project	Project 2	—

Step	Materials	Supplemental (optional)
Supplemental pack	—	Textbook Ch 3.3 — Normal · Ch 4.1–4.2 — Variability & CIs · Khan: confidence intervals · Khan: CI simulation

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Module 3 — Discovering Patterns

Describing a dataset with graphs and summaries.

Step	Materials	Supplemental (optional)
Read 0	Module 3 Read 0	—
Read	Module 3 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 3 Read 2	—
Quiz B	Concept Check	—
Demo lab	Module 3 demo	—
Exercise lab	Module 3 exercise lab	—
Milestone	Canvas Module 3	—
Synthesis	Module 3 synthesis	—
Project	Project 3	—
Supplemental pack	—	Textbook Ch 1 — Intro to data (§1.4–1.7) · R4DS: Data visualization · Posit: ggplot2 video

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Module 4 — Designing Fair Comparisons

Comparing two groups responsibly.

Step	Materials	Supplemental (optional)
Read 0	Module 4 Read 0	—
Read	Module 4 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 4 Read 2	—
Quiz B	Concept Check	—
Demo lab	Module 4 demo	—
Exercise lab	Module 4 exercise lab	—
Milestone	Project 4 milestone	—
Synthesis	Module 4 synthesis	—
Project	Project 4	—
Supplemental pack	—	Textbook Ch 5.3 — Two-sample tests · Ch 8.1–8.2 — Proportions · R4DS: Data transformation

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Module 5 — Looking Ahead

Using data to make predictions.

Step	Materials	Supplemental (optional)
Read 0	Module 5 Read 0	—
Read	Module 5 Read 1	Regression line applet
Quiz A	Def & Notation	—
Read more	Module 5 Read 2	Penn State: prediction intervals
Quiz B	Concept Check	—
Demo lab	Module 5 demo	—
Exercise lab	Module 5 exercise lab	—
Synthesis	Module 5 synthesis	—
Milestone	Canvas Module 5	—
Project	Project 5 · **Portfolio package due Thu Jul 2**	—
Supplemental pack	—	Textbook Ch 6 — Simple linear regression · Khan: regression line example · Penn State: prediction intervals